

Buzzer Beaters

Client: Alexandre Klaus

Team: 13

Date: April 9th, 2026

Time: 3:20 p.m.



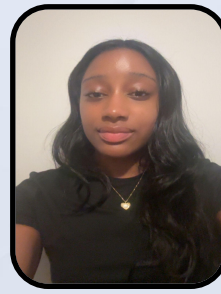
**Mingxuan
(Merrick)
Wang**
Team
Leader



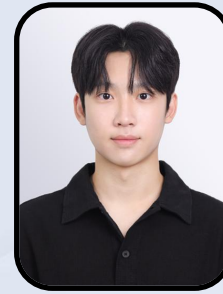
**Shannon
Ho**
Project
Manager



**Omotayo(Omo)
Faseemo**
Communication
Manager



**Kamilia
Desroches
Parchment**
Meeting
Coordinator



**Henry
Kim**
Quality
Assurance
Manager



**Abdulrahman
(AB) Abuloghod**
Research
Coordinator

DISCLAIMER

This design has been prepared by first year engineering students and architects students at the University of Toronto and does not present a Professional Engineering design or Architecture design. A Professional Engineer or Architect has not reviewed this presentation for technical accuracy or adequacy. The recommendations of this design may not be implemented in any way unless reviewed and approved by a licensed Professional Engineer or Architect where such review and approval is required by professional or legal standards, it being understood that it is the responsibility of the recipient of the design to assess whether such a requirement exists. The Presentation may not be reproduced, in whole or in part, without this Disclaimer.

© Engineering Strategies and Practice, University of Toronto, 2026

Permission to reproduce material from this presentation may be requested by email to your team's contact person, with a CC to Engineering Strategies and Practice at esp@engineering.utoronto.ca



Figure 1. Skule™ Kup Logo

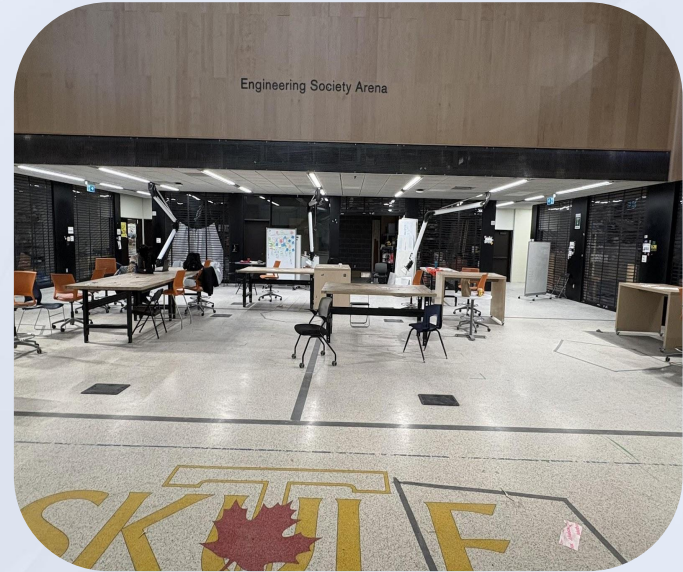


Figure 2. Engineering Society Arena

Omo



Problem

Proposed Design

Key Objectives

Iteration

Next Steps

The Skule™ Wand is an effective, feasible, and efficient design

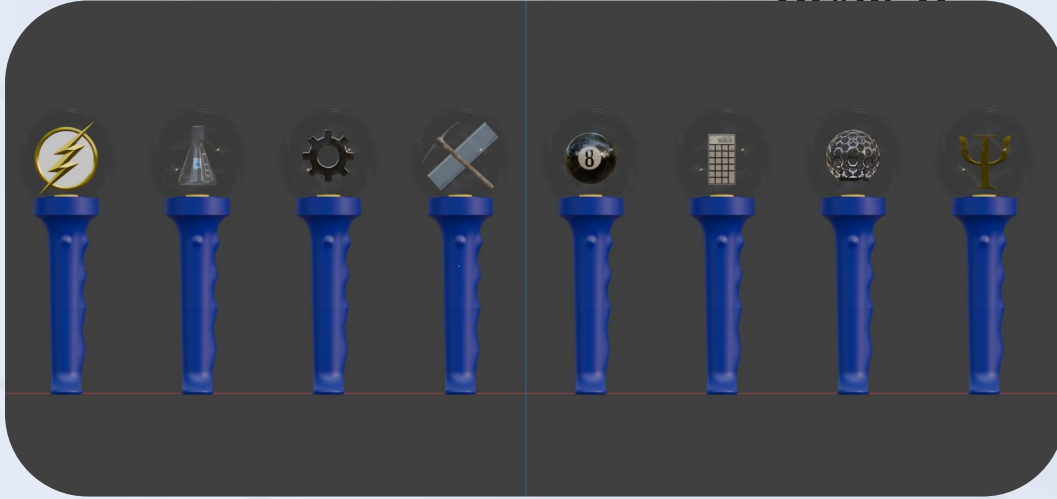


Figure 3. Skule™ Wand 3D Model

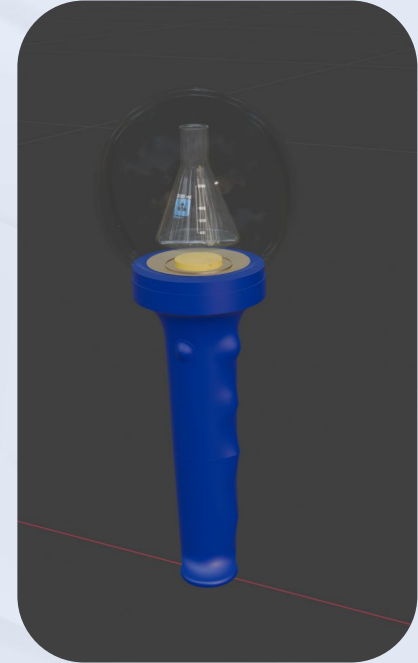


Figure 4. Skule™ Wand Close-up

Shannon



Problem

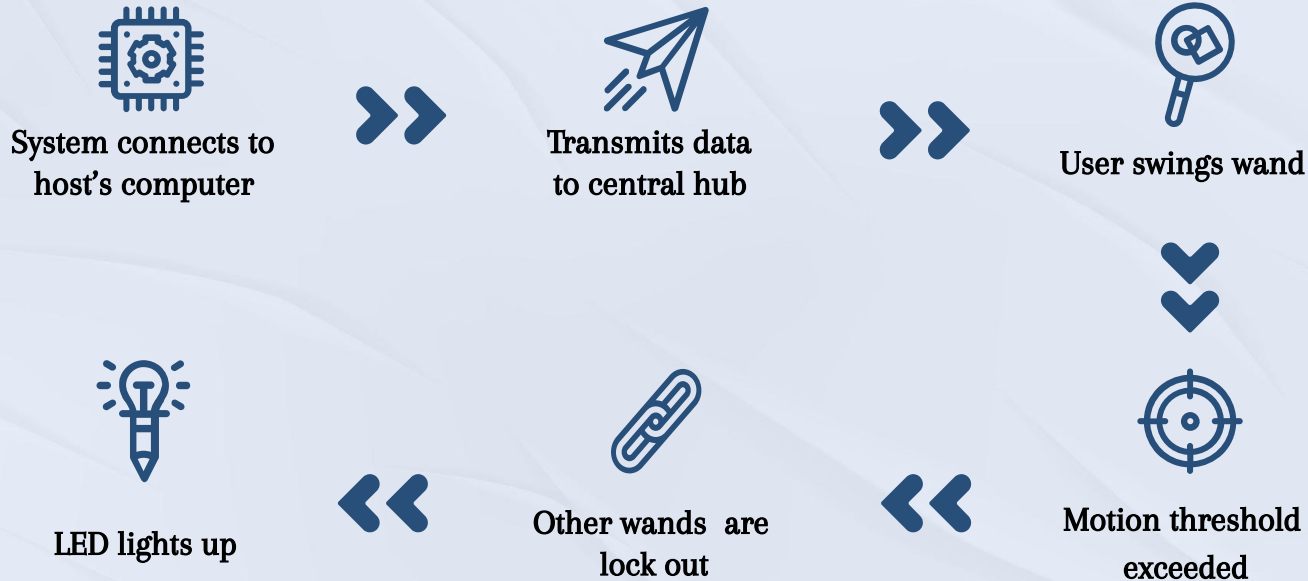
Proposed Design

Key Objectives

Iteration

Next Steps

The Skule™ Wand system identifies the first responder



Henry



Problem

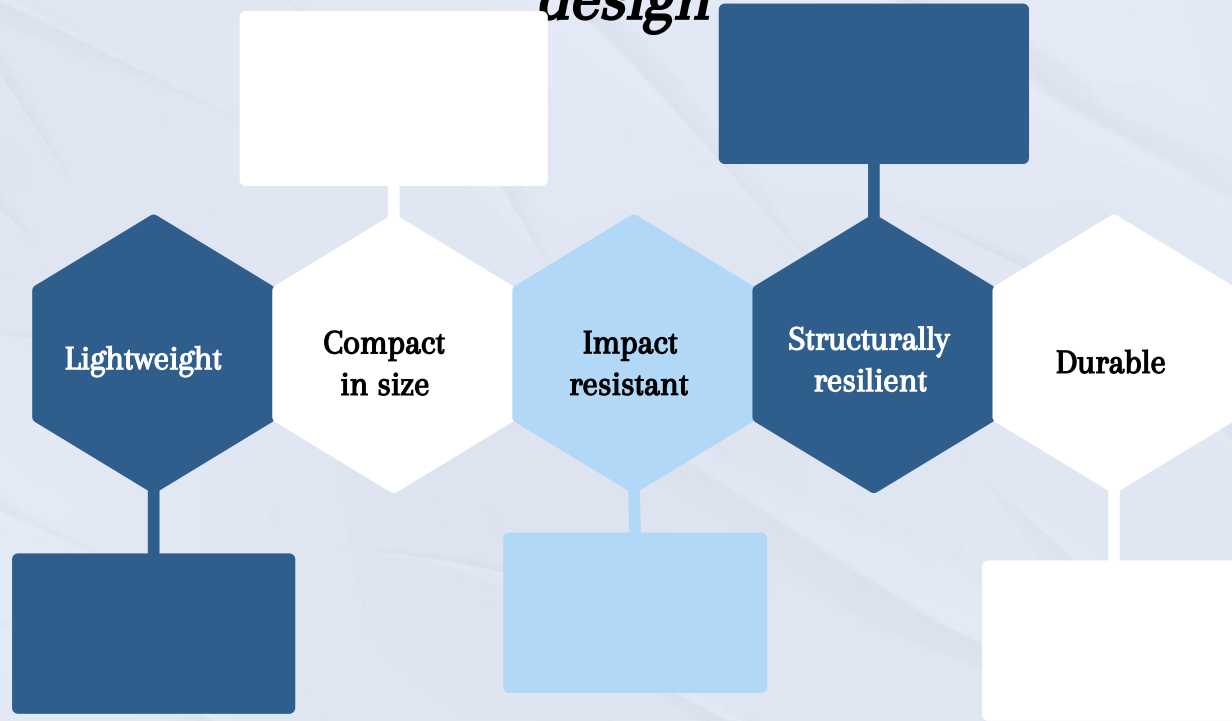
Proposed Design

Key Objectives

Iteration

Next Steps

The Skule™ Wand is an effective, feasible, and efficient design



Kamilia



The Skule™ Wand has been proven to meet all objectives

1. **Lightweight** ✓
2. **Compact in size** ✓
3. **Impact resistant**
4. **Structurally resilient** ✓
5. **Durable** ✓

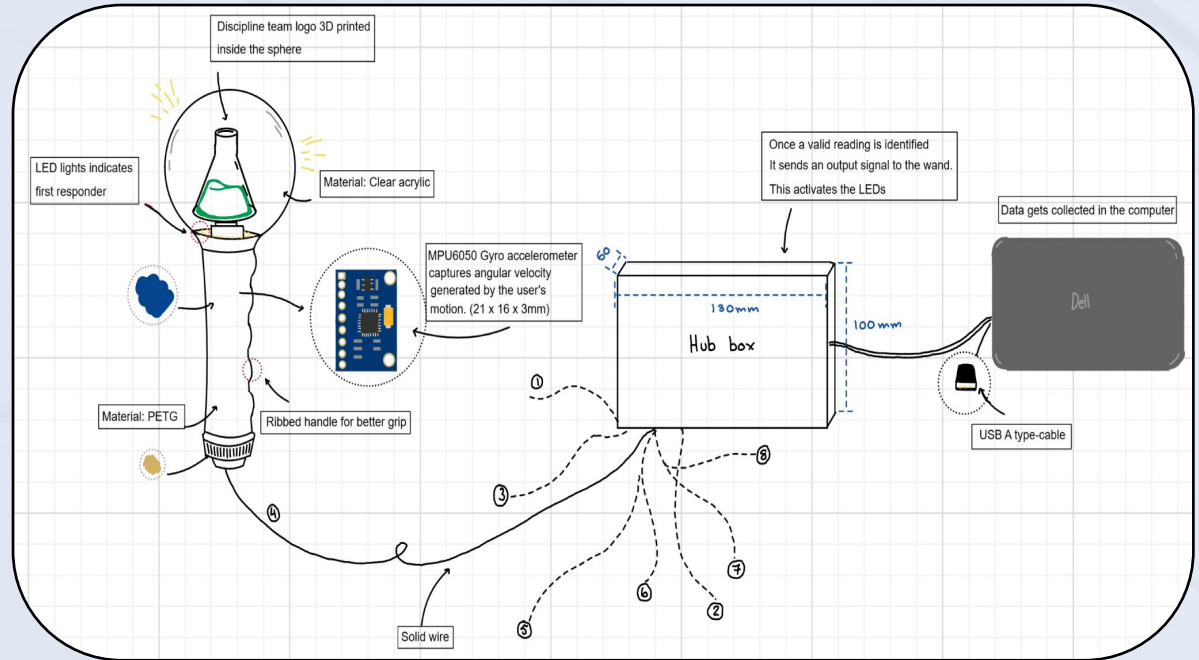


Figure 5. Skule™ Wand System

Kamilia



Problem

Proposed Design

Key Objectives

Iteration

Next Steps

~~MoS~~ Skule™ Wand Skule™ Wand is physically light weight and easily maneuverable



Figure 8. Skule™ Wand Close-up

Impact Resistant (without damage)



Qmo



MoS demonstrates that Skule™ Wand is compact in size

- Meets the Air-Canada carry-on limits ($\leq 250 \times 300 \times 230\text{mm}$)
- System must be portable by one person
- System dimensions: $200 \times 300 \times 230\text{mm}$

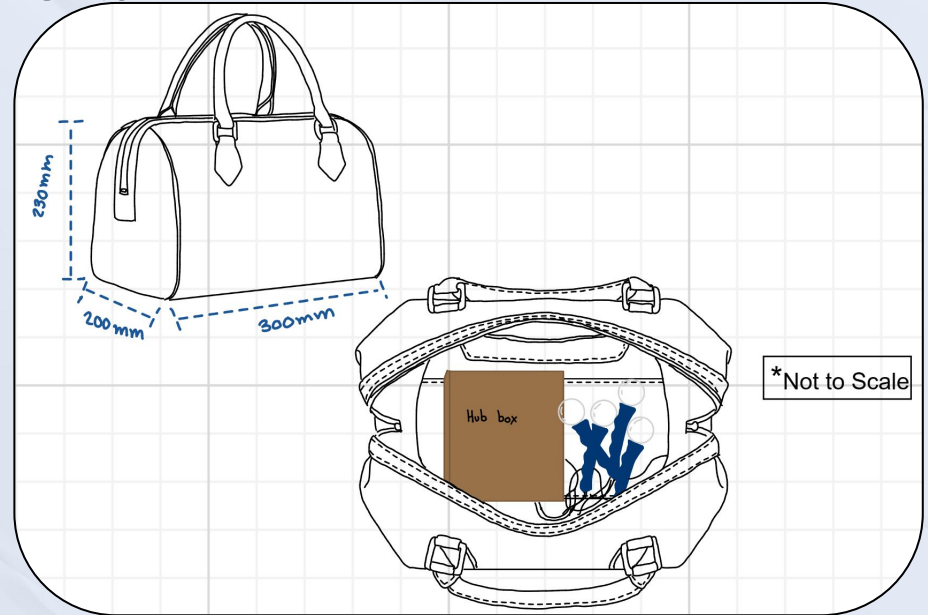


Figure 9. Skule™ Wand System Packaged

AB



Problem

Proposed Design

Key Objectives

Iteration

Next Steps

MoS testing revealed the flaws of the acrylic sphere

Old Material:
Clear Acrylic

Shortcomings : Clear Acrylic sphere
cracks often during drop test



Figure 10. Skule™ Wand Prototype



Figure 11. Broken acrylic sphere

Merrick



Problem

Proposed Design

Key Objectives

Iteration

Next Steps

Clear polycarbonate is more impact resistant than clear acrylic

Previous Design
Model

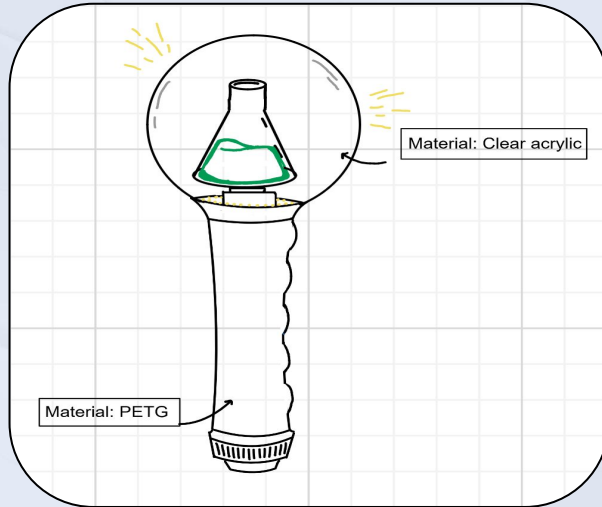


Figure 12. Skule™ Wand Original Design

New Design
Model

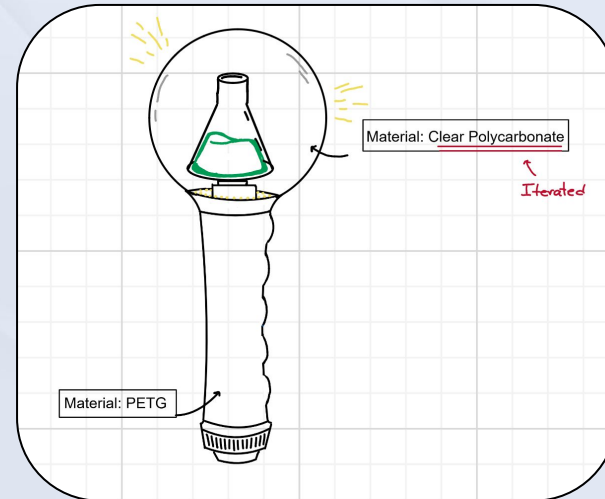


Figure 13. Skule™ Wand New Design

Merrick



Problem

Proposed Design

Key Objectives

Iteration

Next Steps

Clear polycarbonate is more impact resistant than clear acrylic

New Material:
Polycarbonate

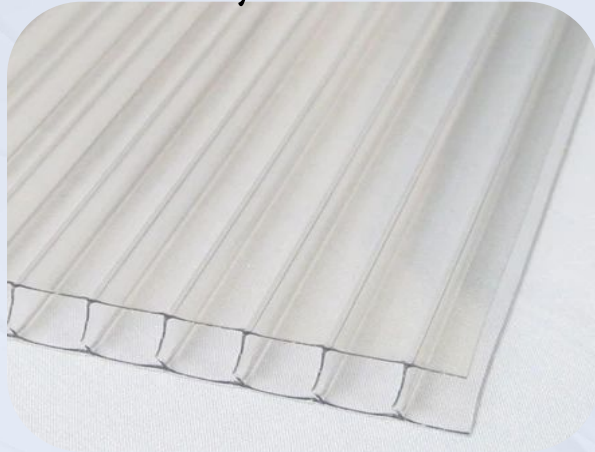


Figure 14. Polycarbonate Sheets [1]

Iteration : Clear acrylic → Clear polycarbonate
Reasoning : Stronger structural resilience

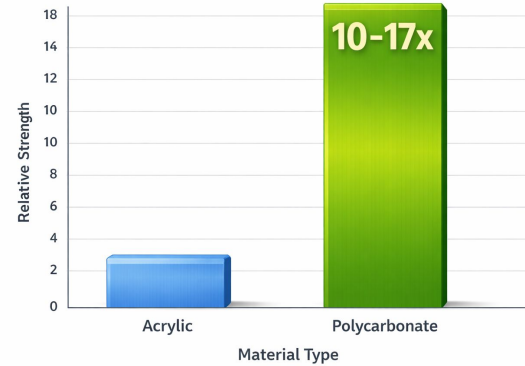


Figure 15. Acrylic and Polycarbonate Strength Comparison [2]

Merrick



Problem

Proposed Design

Key Objectives

Iteration

Next Steps



Next Steps

- Implement the Skule™ Wand in Skule Discipline Feud
- Expand the design's compatibility with different types of hardware connections [3] or Disconnectable/wireless [4]



Figure 16. The Pit

Figure 17. Different Types of Hardware Connections

Henry



Problem

Proposed Design

Key Objectives

Iteration

Next Steps

Takeaway: Our design sufficiently satisfies the client's requirements and is prepared for more comprehensive testing

- Feasible ✓
- Satisfies/exceeds objectives ✓
- Measure of Success (MoS) confirms competency ✓
- Iteration after MoS testing ✓

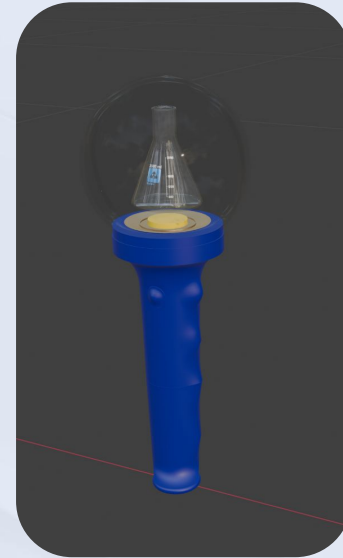


Figure 18. Skule™ Wand Close up

Shannon



Problem

Proposed Design

Key Objectives

Iteration

Next Steps

Thank you for listening

Any questions?



References

[1] Planta Greenhouses, "Triple Wall - Clear 16mm - Thermoclear 15 - Polycarbonate Sheets." Accessed: Apr. 08, 2026. [Online].

Available:<https://plantagreenhouses.com/products/triple-wall-clear-16mm-thermoclear-15-polycarbonate-sheets>

[2] "Polycarbonate vs acrylic - learn about the difference between acrylic & polycarbonate material: A&C plastics," Polycarbonate vs Acrylic - Learn about the Difference Between Acrylic & Polycarbonate Material | A&C Plastics. Accessed: April 8, 2026. [Online].

Available:<https://www.acplasticsinc.com/informationcenter/r/acrylic-vs-polycarbonate>

[3] A comprehensive guide to different types of connectors. Accessed: April 8, 2026. [Online].

Available:<https://seetronic.com/blog/a-comprehensive-guide-to-different-types-of-connectors/>

[4] F. Greer, "6 types of wireless IOT Technology," Zipit, Accessed: April 8, 2026. [Online].

Available:<https://www.zipitwireless.com/blog/5-types-of-wireless-iot-technology>

Threshold logic used in the Skule TMWand system

Step 1: Compute the motion score a from sensor readings

```
float ax = fabs(wand.vx - wand.vxPrior) / dt;
//dt = 0.3 s (Sampling interval)
float ay = fabs(wand.vy - wand.vyPrior) / dt;
float az = fabs(wand.vz - wand.vzPrior) / dt;
float a = sqrt((ax * ax) + (ay * ay) + (az * az));
// ...
wand.a = a;
// ...
float wand1a = wands[0].a;
float wand2a = wands[1].a;
```

Step 2: Compare both wands using the final motion score, a

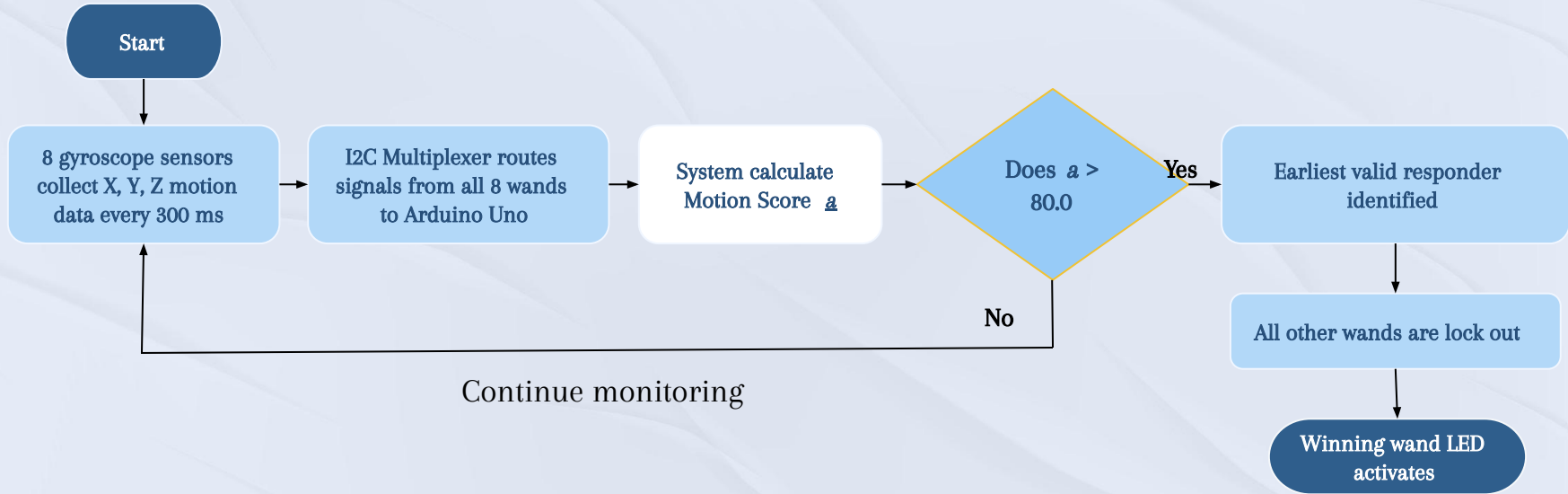
```
float sensitivity = 80.0f; // This is an our 'threshold'

if (wand1a > wand2a && wand1a > sensitivity && turn >=
2){
    // wand 1 wins
}

else if (wand2a > wand1a && wand2a > sensitivity && turn
>= 2){
    // wand 2 wins
}
```

Winner condition: a is higher than the other wand's and $a > 80.0$

How the Skule TM Wand system identifies the first responder



MoS Calculations

$$\text{Success Rate (\%)} = \frac{\text{Number of Successful Trials}}{\text{Total Number of Trials}}$$

Physical Condition

$$\begin{aligned} \text{Success Rate (\%)} &= \frac{32}{36} \times 100\% \\ &= \\ \text{Success Rate (\%)} &= 89\% \end{aligned}$$

Functional Performance

$$\begin{aligned} \text{Success Rate (\%)} &= \frac{33}{36} \times 100\% \\ \bar{\text{Success Rate (\%)}} &= 92\% \end{aligned}$$

Operational Performance

$$\begin{aligned} \text{Success Rate (\%)} &= \frac{30}{36} \times 100\% \\ \bar{\text{Success Rate (\%)}} &= 83\% \end{aligned}$$